Raouf LAARBI

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Profile

Final-year engineering student in Cybersecurity and Cloud Computing at ESILV (degree expected in August 2027) and active bug bounty hunter, passionate about offensive security. Ranked 63th on SignalFisher and 1097th on YesWeHack, with steady progress. Solid experience in application security testing, low-level analysis (reverse engineering, C/C++) and cybersecurity research, notably on intrusion detection in industrial systems (ICS/OT), with ongoing research work. Available from September 2027.

Education

Engineering degree, Cloud Computing & Cybersecurity major — Sept 2023 – Aug 2027 (expected)
ESILV Paris

Preparatory cycle in Artificial Intelligence (CPGE equivalent) — Sept 2022 – June 2023
ENSIA - Algiers

Baccalauréat, Mathematics track — Sept 2021 – June 2022
National Mathematics High School - Algiers

Technical Skills

Offensive security & Bug Bounty: Manual web penetration testing (OWASP Top 10): IDOR, broken access control, XSS, SSRF, injections. Scripting of exploits and dedicated attack tools, writing reproducible reports (impact, PoC, remediation).

Reverse engineering & malware analysis: analysis of binary formats (PE, ELF, DLL, .so), identification of the OEP (`_start`), IAT and entropy analysis, feature extraction.

Systems & detection: Linux internals, Windows Sysinternals, eBPF (Linux kernel), system telemetry, log analysis, Pentesting / Red Teaming.

Programming & automation: Python (advanced, *asyncio*), C/C++, Bash; AI-assisted reconnaissance and triage pipeline to map the attack surface.

AI & applied cryptography: machine learning applied to cybersecurity, *feature engineering*, decentralized federated learning, zk-SNARKs (Circom, snarkjs).

Languages: French, English (B2+).

Research & Security Projects

ICS/OT intrusion detection research — 2025 – Present
University of Tartu (Estonia), InfoSec group

- Study of intrusion detection in industrial control systems (ICS/OT) on real attack datasets (HAI DATASET).
- Wrote a research paper (submitted to NordSec 2026), currently under revision for resubmission.

Research: decentralized federated learning & zk-SNARKs — 2025 – Present

- Co-designed a peer-to-peer federated learning system with no central server, where the integrity of updates is guaranteed by **zk-SNARK** proofs (**Circom**, **snarkjs**) verifying local gradient descent steps.
- Byzantine fault-tolerant consensus protocol (distributed weighted voting) to detect and isolate malicious nodes.
- Research paper written, in final proofreading phase before submission.

Professional Experience

Software & AI Developer (apprenticeship) — July 2024 – Present
De Pardieu Brocas Maffei, Paris

- Designed a RAG (*Retrieval-Augmented Generation*) system based on a fine-tuned **Mistral** LLM, for semantic analysis and querying of sensitive legal documents.

- Pipeline for extraction, analysis and cleaning of **born-digital documents** (PDF, Word), with embeddings stored in a **vector database** for semantic search.
- Developed internal monitoring tools to track infrastructure health and analyze system telemetry.
- Automation of internal processes (scripts) and full-stack web dashboards delivering AI features to users.